

Nombre: ____ Ana Sofía López ____ Matrícula: ____ A0000001 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.6000 < X < 50.2000) = P(-1.2807 < Z < 2.2443) = 0.8875$
3. $X \sim Bin(n = 14, p = 0.05)$
 $P(X = 3) = 0.025880$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.6000 - 49.4000)/(6*0.1702) = 1.1750$
 $C_{pks} = (50.6000 - 49.8180)/(3*0.1702) = 1.5314$
 $C_{pki} = (49.8180 - 49.4000)/(3*0.1702) = 0.8186$
 $C_{pk} = \min(1.5314, 0.8186) = 0.8186$
5. **** Diferencia de medias
 $\nu = 68.3055$
 $LCS = 3.60 - 2.5900 + 1.6676(0.3087) = 1.0100 + 0.5148 = 1.5248$
 $LCI = 3.60 - 2.5900 - 1.6676(0.3087) = 1.0100 - 0.5148 = 0.4952$
** Antes del cambio
 $LCS = 3.6000 + 1.6849(1.0900/6.3246^{**0.5}) = 3.6000 + 0.2904 = 3.8904$
 $LCI = 3.6000 - 1.6849(1.0900/6.3246^{**0.5}) = 3.6000 - 0.2904 = 3.3096$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCS = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0930$
 $\mu = 500 - 8(-1.3225) = 510.5800$

Nombre: ____ Carlos Eduardo García ____ Matrícula: ____ A0000002 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.9000 < X < 50.4000) = P(0.4818 < Z < 3.4193) = 0.3147$
3. $X \sim Bin(n = 15, p = 0.05)$
 $P(X = 4) = 0.004853$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.6000 - 49.4000)/(6*0.1702) = 1.1750$
 $C_{pks} = (50.6000 - 49.8180)/(3*0.1702) = 1.5314$
 $C_{pki} = (49.8180 - 49.4000)/(3*0.1702) = 0.8186$
 $C_{pk} = \min(1.5314, 0.8186) = 0.8186$
5. **** Diferencia de medias
 $\nu = 68.3055$
 $LCS = 3.60 - 3.1400 + 1.9127(0.3087) = 0.4600 + 0.5905 = 1.0505$
 $LCI = 3.60 - 3.1400 - 1.9127(0.3087) = 0.4600 - 0.5905 = -0.1305$
** Antes del cambio
 $LCS = 3.6000 + 1.9371(1.0900/6.3246^{**0.5}) = 3.6000 + 0.3338 = 3.9338$
 $LCI = 3.6000 - 1.9371(1.0900/6.3246^{**0.5}) = 3.6000 - 0.3338 = 3.2662$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCI = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0460$
 $\mu = 500 - 8(-1.6849) = 513.4795$

Nombre: ____ María Fernanda Ruiz ____ Matrícula: ____ A0000003 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.8000 < X < 50.3000) = P(-0.1058 < Z < 2.8317) = 0.5398$
3. $X \sim Bin(n = 15, p = 0.05)$
 $P(X = 7) = 0.000003$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.6000 - 49.4000)/(6*0.1702) = 1.1750$
 $C_{pk} = (50.6000 - 49.8180)/(3*0.1702) = 1.5314$
 $C_{pk} = (49.8180 - 49.4000)/(3*0.1702) = 0.8186$
 $C_{pk} = \min(1.5314, 0.8186) = 0.8186$
5. **** Diferencia de medias
 $\nu = 70.0569$
 $LCS = 3.60 - 2.7600 + 1.8401(0.3049) = 0.8400 + 0.5611 = 1.4011$
 $LCI = 3.60 - 2.7600 - 1.8401(0.3049) = 0.8400 - 0.5611 = 0.2789$
** Antes del cambio
 $LCS = 3.6000 + 1.8617(1.0900/6.4031^{**0.5}) = 3.6000 + 0.3169 = 3.9169$
 $LCI = 3.6000 - 1.8617(1.0900/6.4031^{**0.5}) = 3.6000 - 0.3169 = 3.2831$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCS = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1110$
 $\mu = 500 - 8(-1.2212) = 509.7698$

Nombre: ____ José Manuel Torres ____ Matrícula: ____ A0000004 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(50.0000 < X < 50.1000) = P(1.0693 < Z < 1.6568) = 0.0937$
3. $X \sim Bin(n = 15, p = 0.05)$
 $P(X = 4) = 0.004853$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.5000 - 49.5000)/(6*0.1702) = 0.9792$
 $C_{pks} = (50.5000 - 49.8180)/(3*0.1702) = 1.3356$
 $C_{pki} = (49.8180 - 49.5000)/(3*0.1702) = 0.6227$
 $C_{pk} = \min(1.3356, 0.6227) = 0.6227$
5. **** Diferencia de medias
 $\nu = 82.3169$
 $LCS = 3.60 - 2.6800 + 1.9072(0.2818) = 0.9200 + 0.5375 = 1.4575$
 $LCI = 3.60 - 2.6800 - 1.9072(0.2818) = 0.9200 - 0.5375 = 0.3825$
** Antes del cambio
 $LCS = 3.6000 + 1.9273(1.0900/6.9282^{**0.5}) = 3.6000 + 0.3032 = 3.9032$
 $LCI = 3.6000 - 1.9273(1.0900/6.9282^{**0.5}) = 3.6000 - 0.3032 = 3.2968$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCS = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1130$
 $\mu = 500 - 8(-1.2107) = 509.6858$

Nombre: ____ Daniela Hernández ____ Matrícula: ____ A0000005 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(50.0000 < X < 50.0000) = P(1.0693 < Z < 1.0693) = 0.0000$
3. $X \sim Bin(n = 16, p = 0.05)$
 $P(X = 5) = 0.000776$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.6000 - 49.4000)/(6*0.1702) = 1.1750$
 $C_{pks} = (50.6000 - 49.8180)/(3*0.1702) = 1.5314$
 $C_{pki} = (49.8180 - 49.4000)/(3*0.1702) = 0.8186$
 $C_{pk} = \min(1.5314, 0.8186) = 0.8186$
5. **** Diferencia de medias
 $\nu = 80.5654$
 $LCS = 3.60 - 3.0000 + 2.3739(0.2848) = 0.6000 + 0.6761 = 1.2761$
 $LCI = 3.60 - 3.0000 - 2.3739(0.2848) = 0.6000 - 0.6761 = -0.0761$
** Antes del cambio
 $LCS = 3.6000 + 2.4102(1.0900/6.8557^{**0.5}) = 3.6000 + 0.3832 = 3.9832$
 $LCI = 3.6000 - 2.4102(1.0900/6.8557^{**0.5}) = 3.6000 - 0.3832 = 3.2168$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCS = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1420$
 $\mu = 500 - 8(-1.0714) = 508.5710$

Nombre: ____ Luis Ángel Castro ____ Matrícula: ____ A0000006 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.7000 < X < 50.4000) = P(-0.6932 < Z < 3.4193) = 0.7556$
3. $X \sim Bin(n = 17, p = 0.05)$
 $P(X = 6) = 0.000110$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.6000 - 49.4000)/(6*0.1702) = 1.1750$
 $C_{pk} = (50.6000 - 49.8180)/(3*0.1702) = 1.5314$
 $C_{pk} = (49.8180 - 49.4000)/(3*0.1702) = 0.8186$
 $C_{pk} = \min(1.5314, 0.8186) = 0.8186$
5. **** Diferencia de medias
 $\nu = 77.0626$
 $LCS = 3.60 - 3.5300 + 1.9913(0.2911) = 0.0700 + 0.5796 = 0.6496$
 $LCI = 3.60 - 3.5300 - 1.9913(0.2911) = 0.0700 - 0.5796 = -0.5096$
** Antes del cambio
 $LCS = 3.6000 + 2.0154(1.0900/6.7082^{**0.5}) = 3.6000 + 0.3275 = 3.9275$
 $LCI = 3.6000 - 2.0154(1.0900/6.7082^{**0.5}) = 3.6000 - 0.3275 = 3.2725$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCS = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0440$
 $\mu = 500 - 8(-1.7060) = 513.6483$

Nombre: ____ Valeria Mendoza ____ Matrícula: ____ A0000007 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.9000 < X < 50.4000) = P(0.4818 < Z < 3.4193) = 0.3147$
3. $X \sim Bin(n = 15, p = 0.05)$
 $P(X = 4) = 0.004853$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.5000 - 49.5000)/(6*0.1702) = 0.9792$
 $C_{pks} = (50.5000 - 49.8180)/(3*0.1702) = 1.3356$
 $C_{pki} = (49.8180 - 49.5000)/(3*0.1702) = 0.6227$
 $C_{pk} = \min(1.3356, 0.6227) = 0.6227$
5. **** Diferencia de medias
 $\nu = 66.5541$
 $LCS = 3.60 - 2.8800 + 1.6683(0.3127) = 0.7200 + 0.5216 = 1.2416$
 $LCI = 3.60 - 2.8800 - 1.6683(0.3127) = 0.7200 - 0.5216 = 0.1984$
** Antes del cambio
 $LCS = 3.6000 + 1.6860(1.0900/6.2450^{**0.5}) = 3.6000 + 0.2943 = 3.8943$
 $LCI = 3.6000 - 1.6860(1.0900/6.2450^{**0.5}) = 3.6000 - 0.2943 = 3.3057$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCI = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1590$
 $\mu = 500 - 8(-0.9986) = 507.9886$

Nombre: ____ Diego Alejandro Vega ____ Matrícula: ____ A0000008 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.7000 < X < 50.1000) = P(-0.6932 < Z < 1.6568) = 0.7071$
3. $X \sim Bin(n = 15, p = 0.05)$
 $P(X = 4) = 0.004853$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.5000 - 49.5000)/(6*0.1702) = 0.9792$
 $C_{pk} = (50.5000 - 49.8180)/(3*0.1702) = 1.3356$
 $C_{pk} = (49.8180 - 49.5000)/(3*0.1702) = 0.6227$
 $C_{pk} = \min(1.3356, 0.6227) = 0.6227$
5. **** Diferencia de medias
 $\nu = 77.0626$
 $LCS = 3.60 - 3.4500 + 1.7741(0.2911) = 0.1500 + 0.5164 = 0.6664$
 $LCI = 3.60 - 3.4500 - 1.7741(0.2911) = 0.1500 - 0.5164 = -0.3664$
** Antes del cambio
 $LCS = 3.6000 + 1.7921(1.0900/6.7082^{**0.5}) = 3.6000 + 0.2912 = 3.8912$
 $LCI = 3.6000 - 1.7921(1.0900/6.7082^{**0.5}) = 3.6000 - 0.2912 = 3.3088$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCS = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1570$
 $\mu = 500 - 8(-1.0069) = 508.0549$

Nombre: ____ Fernanda Cárdenas ____ Matrícula: ____ A0000009 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.6000 < X < 50.4000) = P(-1.2807 < Z < 3.4193) = 0.8995$
3. $X \sim Bin(n = 13, p = 0.05)$
 $P(X = 6) = 0.000019$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.6000 - 49.4000)/(6*0.1702) = 1.1750$
 $C_{pk} = (50.6000 - 49.8180)/(3*0.1702) = 1.5314$
 $C_{pk} = (49.8180 - 49.4000)/(3*0.1702) = 0.8186$
 $C_{pk} = \min(1.5314, 0.8186) = 0.8186$
5. **** Diferencia de medias
 $\nu = 71.8083$
 $LCS = 3.60 - 2.9300 + 2.2146(0.3013) = 0.6700 + 0.6672 = 1.3372$
 $LCI = 3.60 - 2.9300 - 2.2146(0.3013) = 0.6700 - 0.6672 = 0.0028$
** Antes del cambio
 $LCS = 3.6000 + 2.2482(1.0900/6.4807^{**0.5}) = 3.6000 + 0.3781 = 3.9781$
 $LCI = 3.6000 - 2.2482(1.0900/6.4807^{**0.5}) = 3.6000 - 0.3781 = 3.2219$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCS = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1390$
 $\mu = 500 - 8(-1.0848) = 508.6786$

Nombre: ____ Miguel Ángel Ríos ____ Matrícula: ____ A0000010 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(50.0000 < X < 50.0000) = P(1.0693 < Z < 1.0693) = 0.0000$
3. $X \sim Bin(n = 14, p = 0.05)$
 $P(X = 6) = 0.000031$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.5000 - 49.5000)/(6*0.1702) = 0.9792$
 $C_{pks} = (50.5000 - 49.8180)/(3*0.1702) = 1.3356$
 $C_{pki} = (49.8180 - 49.5000)/(3*0.1702) = 0.6227$
 $C_{pk} = \min(1.3356, 0.6227) = 0.6227$
5. **** Diferencia de medias
 $\nu = 71.8083$
 $LCS = 3.60 - 2.9600 + 1.9113(0.3013) = 0.6400 + 0.5759 = 1.2159$
 $LCI = 3.60 - 2.9600 - 1.9113(0.3013) = 0.6400 - 0.5759 = 0.0641$
** Antes del cambio
 $LCS = 3.6000 + 1.9343(1.0900/6.4807^{**0.5}) = 3.6000 + 0.3253 = 3.9253$
 $LCI = 3.6000 - 1.9343(1.0900/6.4807^{**0.5}) = 3.6000 - 0.3253 = 3.2747$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCI = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0450$
 $\mu = 500 - 8(-1.6954) = 513.5632$

Nombre: ____ Andrea Martínez ____ Matrícula: ____ A0000011 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.7000 < X < 50.1000) = P(-0.6932 < Z < 1.6568) = 0.7071$
3. $X \sim Bin(n = 14, p = 0.05)$
 $P(X = 6) = 0.000031$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.5000 - 49.5000)/(6*0.1702) = 0.9792$
 $C_{pks} = (50.5000 - 49.8180)/(3*0.1702) = 1.3356$
 $C_{pki} = (49.8180 - 49.5000)/(3*0.1702) = 0.6227$
 $C_{pk} = \min(1.3356, 0.6227) = 0.6227$
5. **** Diferencia de medias
 $\nu = 77.0626$
 $LCS = 3.60 - 2.7900 + 1.9089(0.2911) = 0.8100 + 0.5556 = 1.3656$
 $LCI = 3.60 - 2.7900 - 1.9089(0.2911) = 0.8100 - 0.5556 = 0.2544$
** Antes del cambio
 $LCS = 3.6000 + 1.9305(1.0900/6.7082^{**0.5}) = 3.6000 + 0.3137 = 3.9137$
 $LCI = 3.6000 - 1.9305(1.0900/6.7082^{**0.5}) = 3.6000 - 0.3137 = 3.2863$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCI = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1580$
 $\mu = 500 - 8(-1.0027) = 508.0217$

Nombre: ____ Ricardo Sánchez ____ Matrícula: ____ A0000012 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.6000 < X < 50.2000) = P(-1.2807 < Z < 2.2443) = 0.8875$
3. $X \sim Bin(n = 16, p = 0.05)$
 $P(X = 5) = 0.000776$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.5000 - 49.5000)/(6*0.1702) = 0.9792$
 $C_{pks} = (50.5000 - 49.8180)/(3*0.1702) = 1.3356$
 $C_{pki} = (49.8180 - 49.5000)/(3*0.1702) = 0.6227$
 $C_{pk} = \min(1.3356, 0.6227) = 0.6227$
5. **** Diferencia de medias
 $\nu = 77.0626$
 $LCS = 3.60 - 2.8000 + 1.7741(0.2911) = 0.8000 + 0.5164 = 1.3164$
 $LCI = 3.60 - 2.8000 - 1.7741(0.2911) = 0.8000 - 0.5164 = 0.2836$
** Antes del cambio
 $LCS = 3.6000 + 1.7921(1.0900/6.7082^{**0.5}) = 3.6000 + 0.2912 = 3.8912$
 $LCI = 3.6000 - 1.7921(1.0900/6.7082^{**0.5}) = 3.6000 - 0.2912 = 3.3088$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCS = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0810$
 $\mu = 500 - 8(-1.3984) = 511.1870$

Nombre: ____ Paola Navarro ____ Matrícula: ____ A0000013 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.9000 < X < 50.0000) = P(0.4818 < Z < 1.0693) = 0.1725$
3. $X \sim Bin(n = 13, p = 0.05)$
 $P(X = 6) = 0.000019$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.5000 - 49.5000)/(6*0.1702) = 0.9792$
 $C_{pk} = (50.5000 - 49.8180)/(3*0.1702) = 1.3356$
 $C_{pk} = (49.8180 - 49.5000)/(3*0.1702) = 0.6227$
 $C_{pk} = \min(1.3356, 0.6227) = 0.6227$
5. **** Diferencia de medias
 $\nu = 77.0626$
 $LCS = 3.60 - 3.1300 + 1.8375(0.2911) = 0.4700 + 0.5348 = 1.0048$
 $LCI = 3.60 - 3.1300 - 1.8375(0.2911) = 0.4700 - 0.5348 = -0.0648$
** Antes del cambio
 $LCS = 3.6000 + 1.8571(1.0900/6.7082^{**0.5}) = 3.6000 + 0.3018 = 3.9018$
 $LCI = 3.6000 - 1.8571(1.0900/6.7082^{**0.5}) = 3.6000 - 0.3018 = 3.2982$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCS = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0420$
 $\mu = 500 - 8(-1.7279) = 513.8235$

Nombre: ____ Javier Ramírez ____ Matrícula: ____ A0000014 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.6000 < X < 50.3000) = P(-1.2807 < Z < 2.8317) = 0.8975$
3. $X \sim Bin(n = 13, p = 0.05)$
 $P(X = 4) = 0.002816$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.5000 - 49.5000)/(6*0.1702) = 0.9792$
 $C_{pks} = (50.5000 - 49.8180)/(3*0.1702) = 1.3356$
 $C_{pki} = (49.8180 - 49.5000)/(3*0.1702) = 0.6227$
 $C_{pk} = \min(1.3356, 0.6227) = 0.6227$
5. **** Diferencia de medias
 $\nu = 85.8197$
 $LCS = 3.60 - 3.0200 + 2.3710(0.2761) = 0.5800 + 0.6547 = 1.2347$
 $LCI = 3.60 - 3.0200 - 2.3710(0.2761) = 0.5800 - 0.6547 = -0.0747$
** Antes del cambio
 $LCS = 3.6000 + 2.4049(1.0900/7.0711^{**0.5}) = 3.6000 + 0.3707 = 3.9707$
 $LCI = 3.6000 - 2.4049(1.0900/7.0711^{**0.5}) = 3.6000 - 0.3707 = 3.2293$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCI = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1790$
 $\mu = 500 - 8(-0.9192) = 507.3535$

Nombre: ____ Karla Jiménez ____ Matrícula: ____ A0000015 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.6000 < X < 50.0000) = P(-1.2807 < Z < 1.0693) = 0.7574$
3. $X \sim Bin(n = 16, p = 0.05)$
 $P(X = 7) = 0.000006$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.6000 - 49.4000)/(6*0.1702) = 1.1750$
 $C_{pks} = (50.6000 - 49.8180)/(3*0.1702) = 1.5314$
 $C_{pki} = (49.8180 - 49.4000)/(3*0.1702) = 0.8186$
 $C_{pk} = \min(1.5314, 0.8186) = 0.8186$
5. **** Diferencia de medias
 $\nu = 78.8140$
 $LCS = 3.60 - 3.2900 + 1.8371(0.2879) = 0.3100 + 0.5289 = 0.8389$
 $LCI = 3.60 - 3.2900 - 1.8371(0.2879) = 0.3100 - 0.5289 = -0.2189$
** Antes del cambio
 $LCS = 3.6000 + 1.8561(1.0900/6.7823^{**0.5}) = 3.6000 + 0.2983 = 3.8983$
 $LCI = 3.6000 - 1.8561(1.0900/6.7823^{**0.5}) = 3.6000 - 0.2983 = 3.3017$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCS = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1390$
 $\mu = 500 - 8(-1.0848) = 508.6786$

Nombre: ____ Sebastián Flores ____ Matrícula: ____ A0000016 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.8000 < X < 50.4000) = P(-0.1058 < Z < 3.4193) = 0.5418$
3. $X \sim Bin(n = 15, p = 0.05)$
 $P(X = 6) = 0.000049$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.4000 - 49.6000)/(6*0.1702) = 0.7833$
 $C_{pks} = (50.4000 - 49.8180)/(3*0.1702) = 1.1398$
 $C_{pki} = (49.8180 - 49.6000)/(3*0.1702) = 0.4269$
 $C_{pk} = \min(1.1398, 0.4269) = 0.4269$
5. **** Diferencia de medias
 $\nu = 66.5541$
 $LCS = 3.60 - 2.7500 + 1.9966(0.3127) = 0.8500 + 0.6242 = 1.4742$
 $LCI = 3.60 - 2.7500 - 1.9966(0.3127) = 0.8500 - 0.6242 = 0.2258$
** Antes del cambio
 $LCS = 3.6000 + 2.0244(1.0900/6.2450^{**0.5}) = 3.6000 + 0.3533 = 3.9533$
 $LCI = 3.6000 - 2.0244(1.0900/6.2450^{**0.5}) = 3.6000 - 0.3533 = 3.2467$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCS = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0800$
 $\mu = 500 - 8(-1.4051) = 511.2406$

Nombre: ____ Natalia Moreno ____ Matrícula: ____ A0000017 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.7000 < X < 50.0000) = P(-0.6932 < Z < 1.0693) = 0.6134$
3. $X \sim Bin(n = 15, p = 0.05)$
 $P(X = 6) = 0.000049$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.6000 - 49.4000)/(6*0.1702) = 1.1750$
 $C_{pk} = (50.6000 - 49.8180)/(3*0.1702) = 1.5314$
 $C_{pk} = (49.8180 - 49.4000)/(3*0.1702) = 0.8186$
 $C_{pk} = \min(1.5314, 0.8186) = 0.8186$
5. **** Diferencia de medias
 $\nu = 64.8026$
 $LCS = 3.60 - 3.5300 + 2.2195(0.3167) = 0.0700 + 0.7030 = 0.7730$
 $LCI = 3.60 - 3.5300 - 2.2195(0.3167) = 0.0700 - 0.7030 = -0.6330$
** Antes del cambio
 $LCS = 3.6000 + 2.2570(1.0900/6.1644^{**0.5}) = 3.6000 + 0.3991 = 3.9991$
 $LCI = 3.6000 - 2.2570(1.0900/6.1644^{**0.5}) = 3.6000 - 0.3991 = 3.2009$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCS = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1020$
 $\mu = 500 - 8(-1.2702) = 510.1619$

Nombre: ____ Alejandro Ortiz ____ Matrícula: ____ A0000018 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.9000 < X < 50.2000) = P(0.4818 < Z < 2.2443) = 0.3026$
3. $X \sim Bin(n = 16, p = 0.05)$
 $P(X = 4) = 0.006147$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.7000 - 49.3000)/(6*0.1702) = 1.3708$
 $C_{pks} = (50.7000 - 49.8180)/(3*0.1702) = 1.7273$
 $C_{pki} = (49.8180 - 49.3000)/(3*0.1702) = 1.0144$
 $C_{pk} = \min(1.7273, 1.0144) = 1.0144$
5. **** Diferencia de medias
 $\nu = 61.2998$
 $LCS = 3.60 - 3.2800 + 1.7803(0.3254) = 0.3200 + 0.5794 = 0.8994$
 $LCI = 3.60 - 3.2800 - 1.7803(0.3254) = 0.3200 - 0.5794 = -0.2594$
** Antes del cambio
 $LCS = 3.6000 + 1.8030(1.0900/6.0000^{**0.5}) = 3.6000 + 0.3275 = 3.9275$
 $LCI = 3.6000 - 1.8030(1.0900/6.0000^{**0.5}) = 3.6000 - 0.3275 = 3.2725$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCI = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1590$
 $\mu = 500 - 8(-0.9986) = 507.9886$

Nombre: ____ Camila Salazar ____ Matrícula: ____ A0000019 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.7000 < X < 50.2000) = P(-0.6932 < Z < 2.2443) = 0.7435$
3. $X \sim Bin(n = 16, p = 0.05)$
 $P(X = 7) = 0.000006$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.4000 - 49.6000)/(6*0.1702) = 0.7833$
 $C_{pks} = (50.4000 - 49.8180)/(3*0.1702) = 1.1398$
 $C_{pki} = (49.8180 - 49.6000)/(3*0.1702) = 0.4269$
 $C_{pk} = \min(1.1398, 0.4269) = 0.4269$
5. **** Diferencia de medias
 $\nu = 59.5484$
 $LCS = 3.60 - 3.2100 + 2.3912(0.3300) = 0.3900 + 0.7892 = 1.1792$
 $LCI = 3.60 - 3.2100 - 2.3912(0.3300) = 0.3900 - 0.7892 = -0.3992$
** Antes del cambio
 $LCS = 3.6000 + 2.4411(1.0900/5.9161^{**0.5}) = 3.6000 + 0.4498 = 4.0498$
 $LCI = 3.6000 - 2.4411(1.0900/5.9161^{**0.5}) = 3.6000 - 0.4498 = 3.1502$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCI = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.0420$
 $\mu = 500 - 8(-1.7279) = 513.8235$

Nombre: ____ Héctor Gómez ____ Matrícula: ____ A0000020 ____

1. $\hat{\sigma} = 0.1518/2.326 = 0.0653$
 $LCS = 2.4933 + 3(0.0653/5^{**0.5}) = 2.4933 + 0.0876 = 2.5809$
 $LCI = 2.4933 - 3(0.0653/5^{**0.5}) = 2.4933 - 0.0876 = 2.4057$
2. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $P(49.9000 < X < 50.3000) = P(0.4818 < Z < 2.8317) = 0.3127$
3. $X \sim Bin(n = 16, p = 0.05)$
 $P(X = 4) = 0.006147$
4. $\hat{\sigma} = 0.1920/1.128 = 0.1702$
 $C_p = (50.5000 - 49.5000)/(6*0.1702) = 0.9792$
 $C_{pks} = (50.5000 - 49.8180)/(3*0.1702) = 1.3356$
 $C_{pki} = (49.8180 - 49.5000)/(3*0.1702) = 0.6227$
 $C_{pk} = \min(1.3356, 0.6227) = 0.6227$
5. **** Diferencia de medias
 $\nu = 71.8083$
 $LCS = 3.60 - 2.6400 + 1.9113(0.3013) = 0.9600 + 0.5759 = 1.5359$
 $LCI = 3.60 - 2.6400 - 1.9113(0.3013) = 0.9600 - 0.5759 = 0.3841$
** Antes del cambio
 $LCS = 3.6000 + 1.9343(1.0900/6.4807^{**0.5}) = 3.6000 + 0.3253 = 3.9253$
 $LCI = 3.6000 - 1.9343(1.0900/6.4807^{**0.5}) = 3.6000 - 0.3253 = 3.2747$
6. $\hat{\sigma} = 0.0509/0.9213 = 0.0552$
 $LCS = 0.0509 + 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 + 0.0645 = 0.1154$
 $LCI = 0.0509 - 3((1 - 0.9213^{**2})^{**0.5})0.0552 = 0.0509 - 0.0645 = -0.0136$
7. $P(X < 500) = P(Z < (500 - \mu)/8) = 0.1980$
 $\mu = 500 - 8(-0.8488) = 506.7903$